Def.

Let $f:[a,b]\to\mathbb{C}$ be a function. Given a partition $P=\{x_0,\cdots,x_n\}$, define
$$V(f,P)\overset{\text{def}}{=}\sum_{i=1}^{n}|f(x_i)-f(x_{i-1})|$$
which is called the variation of f with respect to a partition P. And, the value
$$TV(f)\overset{\text{def}}{=}\sup\{V(f,P)\mid P\text{ is partition on }[a,b]\}$$
is called the total variation of f.

If the total variation is finite, we say f is of bounded variation.

Prop.

If $f:[a,b]\to\mathbb{C}$ is continuously differentiable, then f is of bounded variation with
$$TV(f)=\int_a^b|f'(t)|\,dt$$

Proof. Given a partition $P=\{x_0,\cdots,x_n\}$ on $[0,1]$, by the fundamental theorem of calculus,
$$V(f,P)=\sum_{i=1}^{n}|f(x_i)-f(x_{i-1})|=\sum_{i=1}^{n}\left|\int_{x_{i-1}}^{x_i}f'(t)\,dt\right|\leq\sum_{i=1}^{n}\int_{x_{i-1}}^{x_i}|f'(t)|\,dt=\int_a^b|f'(t)|\,dt$$
Since P was arbitrary, $TV(f)\leq\int_a^b|f'(t)|\,dt$, so the total variation has finite value.

To show that the reverse inequality, we use the fact that:
f' is continuous uniformly, because it is continuous on the compact set $[a,b]$.
Given $\varepsilon>0$, take $\delta_1>0$ such that $|x-t|<\delta_1\Rightarrow|f'(x)-f'(t)|$.
And, take $\delta_2>0$ such that if a partition P on $[a,b]$ satisfies $\|P\|<\delta_2$, then
$$\left|\int_a^b|f'(t)|\,dt-\sum_{i=1}^{n}|f'(z_i)|\cdot|x_{i+1}-x_i|\right|<\varepsilon$$
where $P=\{x_0,\cdots,x_n\}$ and $z_i\in[x_i,x_{i+1}]$ is arbitrary. The existence of δ_2 is given by the fact that $|f'|$ is Riemann-integrable.